

Anusha Madasu

Data Engineer

Lubbock,Texas | 806-319-5806 | anu.madasuu@gmail.com | [Linkedin](#)



PROFESSIONAL SUMMARY:

- Over 5 years of IT experience in the field of Data Engineering, analysis, modeling, Development and Project Management
- Expert level in writing **Python, PySpark, SQL, Scala** scripts to build data pipelines and data transformations.
- Extensive experience in working with relational databases like **PostgreSQL, MySQL, SQL Server, AWS RDS, Azure SQL database, Big Query.**
- Hands-on expertise with cloud platforms (AWS, Azure, GCP) and modern data lake house solutions (Databricks, Snowflake, Delta Lake, Apache Iceberg, Hudi).
- Hands on experience in using Hadoop ecosystem components like Hadoop, Hive, Pig, Sqoop, HBase, MapReduce framework, HDFS, Yarn.
- Automated infrastructure provisioning and configuration management using Terraform/ Pulumi and AWS CloudFormation.
- Worked on ETL by developing and deploying AWS Lambda functions for generating a serverless data pipeline which can be written to **Glue** Catalog and can be queried from Athena.
- Adept at implementing real-time streaming solutions (Kafka, Flink, Kinesis), ensuring data quality, governance compliance.
- Experienced in data governance and compliance, ensuring data quality and adherence to regulatory requirements like **GDPR, HIPAA, and CCPA.**
- Working knowledge of data migration, data profiling, data cleaning and transformation utilizing a variety of ETL such as **Apache NiFi and Apache Airflow.**
- Extensive experience in setting up **CI/CD** pipelines using tools such as **Jenkins, Bit Bucket, GitHub, Maven, SVN and Azure DevOps.**
- Good experience in designing and developing logical and physical data models that utilize concepts like **Star Schema, Snowflake Schema,** and Slowly Changing Dimensions.
- Passionate about emerging trends like Data Mesh, AI/ML-driven data pipelines, and vector databases (**Pinecone, Weaviate, FAISS**) for next-gen analytics.
- Integrated AI models with **AWS Sage Maker and GCP Vertex AI** for scalable model training and deployment, ensuring high availability of real-time decision-making capabilities in warehouse robots.
- Knowledge in Data Visualization and analytics using Tableau desktop, Matplotlib, Seaborn, Plotly, and Power BI desktop.
- Proficient in project management and documentation using **JIRA, Prometheus, Grafana, and ELK Stack and JS docs.**
- Experienced in containerization of Applications (**Micro services**) with **Docker and Kubernetes.**

TECHNICAL SKILLS:

- **Programming & Data Processing:** Python, SQL, Scala, Java, Bash
- **Big Data & Streaming:** Apache Spark, Hadoop, Kafka, Flink, Pulsar
- **Cloud Platforms:** AWS (Glue, Redshift, EMR, Kinesis, S3), Azure (ADF, Synapse, Databricks), GCP (BigQuery, Dataflow, Pub/Sub)
- **Data Lakehouse & Warehousing:** Snowflake, Databricks Delta Lake, Apache Iceberg, Hudi
- **Databases:** PostgreSQL, MySQL, MongoDB, Cassandra, DynamoDB, Neo4j
- **Workflow Orchestration:** Apache Airflow, Prefect
- **DevOps for Data:** Docker, Kubernetes, Terraform, Jenkins, GitLab CI/CD
- **Data Governance & Quality:** Great Expectations, Collibra, Alation, Apache Atlas

PROFESSIONAL EXPERIENCE:

Client: UBS Financial Services, NYC

Aug 2024-Till Date

Role: Data Engineer

Responsibilities:

- Involved in building data pipelines to extract transform and load the raw data and perform analytics on the transformed data using AWS services **S3, Glue, Redshift, EMR, RDS, Dynamo DB, Lambda, Athena, KMS** focusing on high-availability, fault tolerance, and auto-scaling using AWS Cloud Formation.
- Built and managed real-time data pipelines using **Apache Kafka, Kafka Streams,** and Kafka Connect for low-latency event processing.
- Developed distributed data processing solutions using **Spark, PySpark, and Spark SQL,** optimizing performance in big

data environments.

- Managed **AWS Glue Data Catalog**, ensuring efficient data ingestion, transformation, and metadata management.
- Orchestrated data workflows with **Apache Airflow**, automating ETL pipelines and enhancing workflow efficiency.
- Designed **AWS Data Lake** architectures, ensuring secure and scalable data storage for structured and unstructured datasets.
- Proficient in ETL processes, utilizing tools like **Apache NiFi, Apache Airflow, dbt, Talend, Informatica Snowflake** for ingesting and processing data from disparate sources.
- Developed and maintained CI/CD pipelines using **Jenkins**, enabling continuous integration, deployment, and infrastructure automation.
- Strong understanding of data governance, lineage, and quality frameworks (Great Expectations, Collibra, Alation) ensuring compliance with **GDPR, HIPAA, and CCPA** standards.
- Facilitated large-scale data migration from HDFS and on-premise systems using Apache Sqoop.
- Designed, deployed, and monitored machine learning models using **Python, PyTorch, and MLflow**, ensuring model lifecycle automation.
- Automated infrastructure provisioning using Infrastructure as Code (IaC) with **AWS CloudFormation** and Ansible.
- Developed and maintained containerized data solutions using **Docker and Kubernetes**, improving scalability and resource efficiency.

Environment: AWS (S3, Glue, Redshift, EMR, RDS, Dynamo DB, Lambda, Athena, KMS), MongoDB, SQL, Python, PySpark, Scala, Airflow, Hive, HDFS, Snowflake, Databricks, Talend, Kubernetes, Tableau, Git.

Client: UHG, IL

Nov 2023 -Jul 2024

Role: Data Engineer

Responsibilities:

- Designed and optimized SQL queries, utilizing complex joins, CTEs, pivots, window functions, and stored procedures for efficient data processing.
- Developed and orchestrated ETL pipelines with **Azure Data Factory (ADF) and DBT (Data Build Tool)** to streamline data ingestion and transformation.
- Modeled scalable data architectures using **Star Schema, Databricks, and Snowflake**, optimizing data structures for analytics and reporting.
- Enforced access control and compliance policies using **Apache Ranger**, Azure Key Vault, and Azure Active Directory (AAD).
- Managed data storage solutions using **ADLS, Hadoop, and Snowflake**, ensuring scalability and cost optimization in Azure Cloud.
- Developed and optimized **Apache Spark and Spark-SQL** workflows, leveraging distributed computing for big data analytics.
- Designed and maintained **NoSQL databases** like Cassandra, ensuring high read/write performance and efficient data retrieval.
- Built machine learning models using TensorFlow and Azure ML, applying feature engineering for predictive analytics.
- Developed batch processing and real-time analytics using **MapReduce, Hive, and Spark**, ensuring efficient data processing.
- Managed workflow orchestration with **Apache Airflow**, automating ETL workflows for streamlined data operations.
- Built and deployed CI/CD pipelines using **Azure DevOps, GitHub, and Terraform**, automating testing, version control, and deployments.

Environment: SQL, ADF, DBT, Star Schema, Databricks, Snowflake, Kubeflow, Apache Ranger, Azure Key Vault, Azure AD, ADLS, Hadoop, Apache Kafka, Apache Spark, Spark-SQL, Cassandra, TensorFlow, Azure ML, MapReduce, Hive, Apache Airflow, Power BI, Grafana, Azure, ServiceNow, Azure DevOps, GitHub, Terraform, REST APIs, JavaScript, UDFs.

Client: HCL Technologies, India

Sep 2019 - Aug 2022

Role: Data Engineer

Responsibilities:

- Migrated traditional ETL processes to a **cloud-based Data Lake** by implementing **Spark** and **Hive jobs**, enhancing data processing efficiency.
- ETL design of the flow and the logic for the Data warehouse project using **Informatica, Shell Scripting**.
- Involved in complete **SDLC** ETL process from development to testing and production environments.

- Orchestrated real-time warehouse management reports, synergizing with automated sorting machinery data, using SSRS and **Tableau**.
- Installed, configured, and maintained Apache Hadoop clusters for application development and major components of Hadoop Ecosystem: **Hive, Pig, HBase, Sqoop, Flume, Oozie and Zookeeper**.
- Worked on creating dashboards in **Tableau** for a collection of several views and comparing the data simultaneously.
- Written Map Reduce program in Python with the Hadoop streaming API.
- Excellent experience in **ETL** analysis, designing, developing, testing and implementing ETL processes including performance tuning and query optimizing of database.
- Perform data scraping, data cleaning, data analysis, and data interpretation and generate meaningful reports using **Python** libraries like **pandas, matplotlib, etc.**
- Used **Maven** as the build tool and SVN for code management.
- Extracted data from flat files and Sql Server and Sybase and **Oracle Database** and applied business logic and loaded in to oracle and **Sybase Warehouse**.

Environment: Hadoop, MapReduce, HDFS, HBase, Hive, Impala, Pig, SQL, Ganglia, Sqoop, Flume, Oozie, Unix, Java, Java Script, Maven, Eclipse, Oracle.

PROFESSIONAL CERTIFICATIONS

Microsoft Certified: Azure Data Scientist Associate – Microsoft

- Comprehensive certification validating **Azure Machine Learning, data preparation, model training, model evaluation, and deployment** workflows.
- Confirms proficiency in **cloud-based ML, MLOps, pipeline automation, and scalable production workloads** on Azure.
- Demonstrates ability to deliver **advanced analytics, machine learning solutions, and real-world business insights** using Microsoft's data science ecosystem.

AWS Certified Data Engineer Associate – Amazon

- Proven ability to design and operationalize end-to-end **data ingestion, transformation, and governance** workflows using **AWS Glue, Lambda, EMR, Redshift, and Lake Formation**.
- Skilled in architecting scalable, secure data lake and **warehouse solutions, optimizing performance, ensuring data quality**, and implementing best-practice governance on AWS.
- Certified in **automating data pipelines and supporting production-grade analytics** systems with monitoring, troubleshooting, and cost-optimized AWS service design.

EDUCATION

Master of Science, Texas Tech University, Texas

May 2024